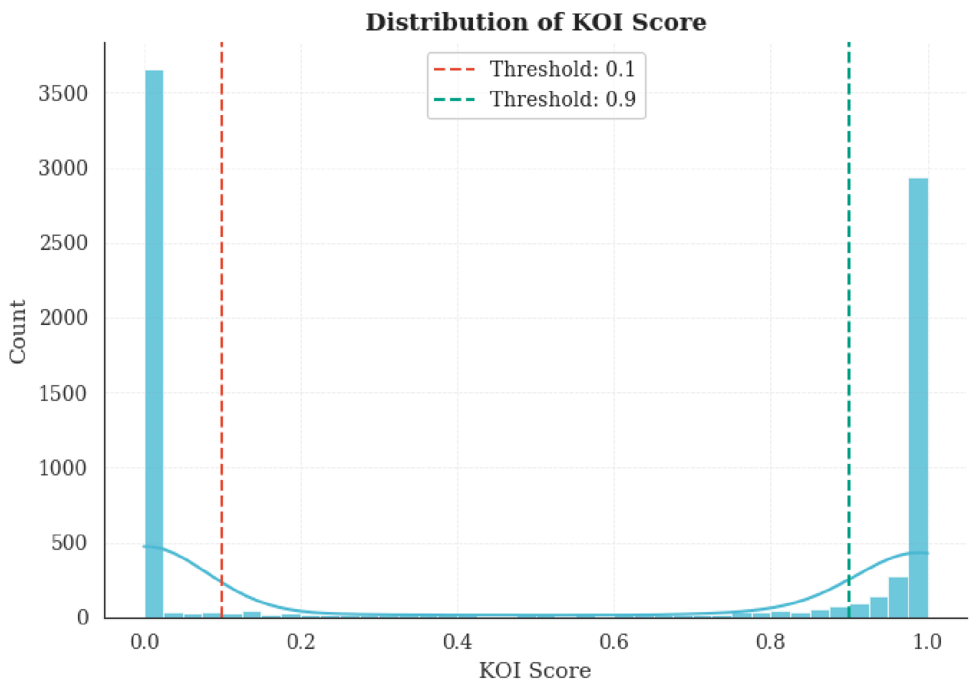
RESULTS

KOI Score Distribution



Distribution_of_KOI_score.png

Summary Statistics

```
count    7994.000000
mean        0.483829
std         0.477009
min         0.000000
25%         0.000000
50%         0.371000
75%         0.999000
max         1.000000
```

After Applying Thresholds

```
Labeled samples: 7211
Label distribution (0=low quality, 1=high quality):
quality_label
0    3753
1    3458
Name: count, dtype: int64
```

Exploratory Data Analysis

Feature Transformations

Log_{10} transformation

- `koi_period`, `koi_duration`, `koi_prad`, `koi_teq`, `koi_insol`, `koi_srad`

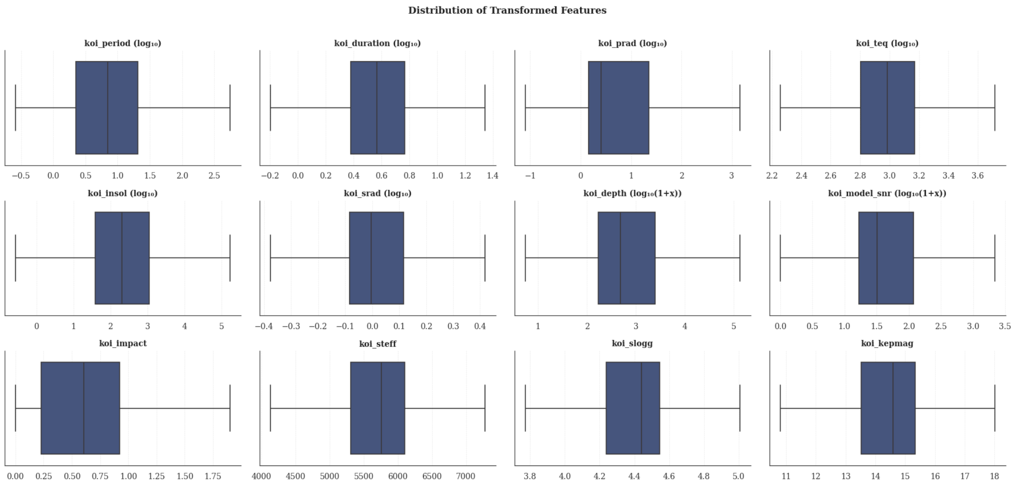
$\text{Log}_{10}(1{+}x)$ transformation

- `koi_depth`, `koi_model_snr`

Linear (no transformation)

- `koi_impact`, `koi_steff`, `koi_slogg`, `koi_kepmag`

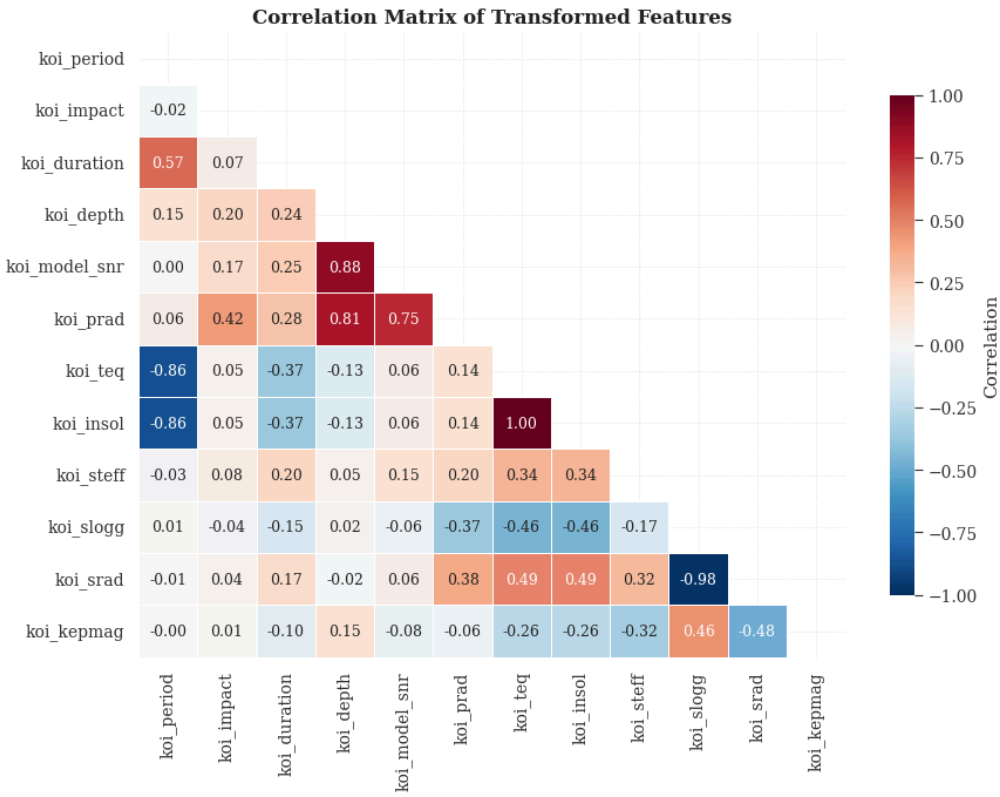
Feature Distribution Boxplots



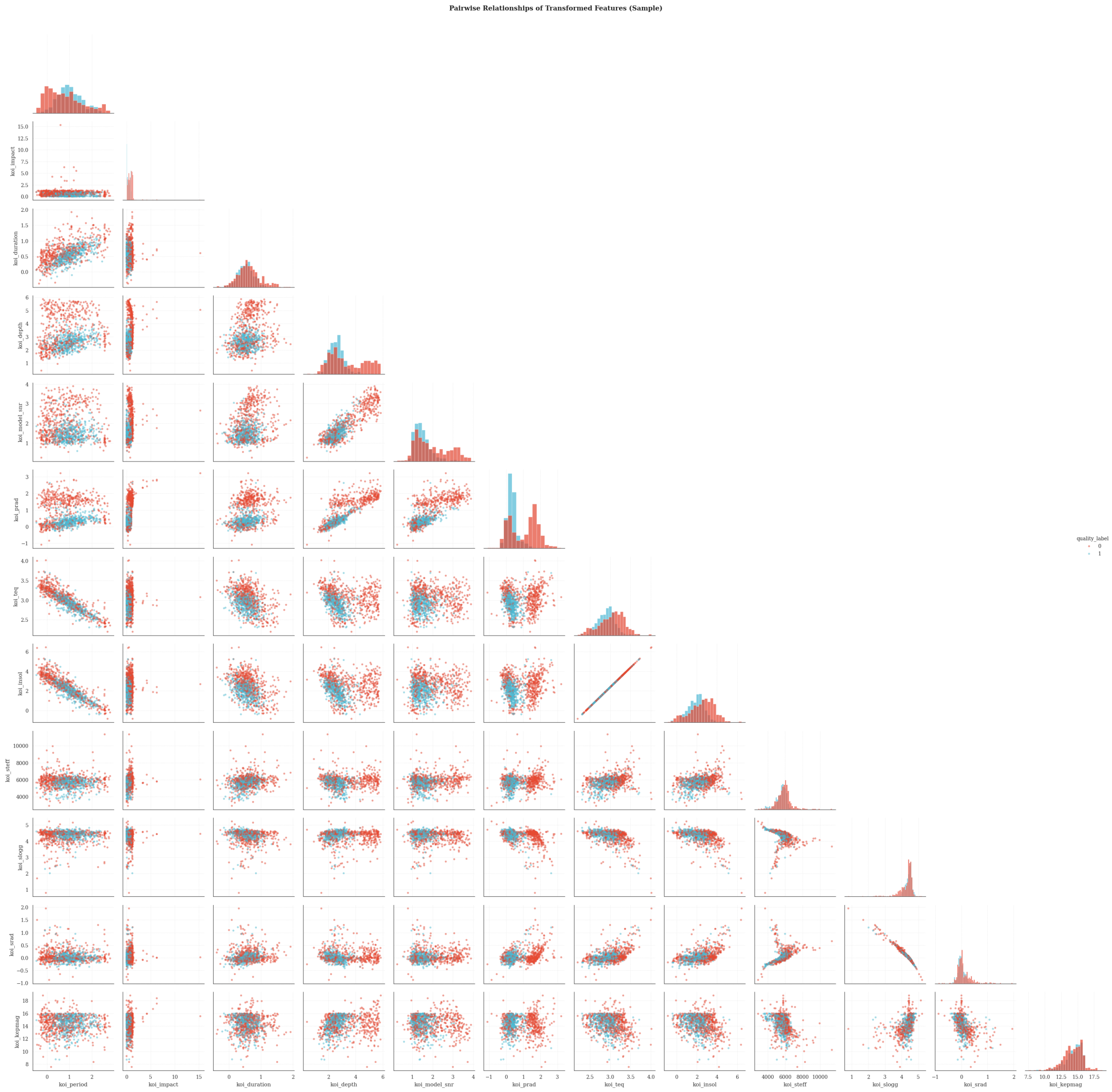
Distribution_of_Transformed_Features.png

Correlation Analysis

Pre-PCA Correlation Matrix (X_{12})



Correlation_Matrix_of_Transformed Features.png



Pairplot_of_Transformed_Features.png

Principal Component Analysis

Eigenvalue Decomposition

Explained variance by component:

	eigenvalue	explained_%	cumulative_%
PC1	3.737318	31.14	31.14
PC2	2.971272	24.76	55.90
PC3	2.111366	17.59	73.49
PC4	0.994332	8.29	81.78
PC5	0.926532	7.72	89.50
PC6	0.646095	5.38	94.88
PC7	0.439769	3.66	98.54
PC8	0.115413	0.96	99.51
PC9	0.052262	0.44	99.94

PC10	0.006205	0.05	99.99
PC11	0.000935	0.01	100.00
PC12	0.000001	0.00	100.00

Key Insight: The first 3 principal components capture 73.49% of the total variance, suggesting strong dimensional reduction potential.

Loading Matrix

Eigenvector matrix A (feature contributions to each PC):

$$\mathbf{A} = \begin{bmatrix} -0.31 & 0.30 & 0.39 & -0.02 & 0.04 & 0.01 & -0.31 & -0.11 & -0.05 & 0.04 & 0.74 & 0.00 \\ 0.08 & 0.19 & -0.16 & -0.09 & 0.92 & -0.15 & 0.06 & -0.20 & 0.11 & 0.00 & 0.00 & 0.00 \\ -0.09 & 0.36 & 0.31 & 0.22 & 0.02 & 0.32 & 0.78 & 0.08 & 0.08 & 0.01 & 0.00 & 0.00 \\ 0.05 & 0.47 & -0.33 & -0.04 & -0.23 & -0.01 & -0.15 & 0.08 & 0.76 & 0.03 & 0.01 & 0.00 \\ 0.15 & 0.43 & -0.31 & 0.11 & -0.28 & -0.22 & 0.11 & -0.62 & -0.41 & -0.01 & -0.01 & 0.00 \\ 0.23 & 0.47 & -0.17 & -0.16 & 0.06 & 0.05 & -0.13 & 0.67 & -0.46 & -0.01 & 0.00 & 0.00 \\ 0.47 & -0.20 & -0.15 & 0.05 & -0.02 & 0.11 & 0.14 & 0.01 & 0.05 & 0.03 & 0.42 & -0.71 \\ 0.47 & -0.20 & -0.15 & 0.05 & -0.02 & 0.11 & 0.14 & 0.01 & 0.05 & 0.03 & 0.42 & 0.71 \\ 0.22 & 0.09 & 0.13 & 0.78 & 0.13 & 0.29 & -0.41 & -0.04 & 0.00 & -0.13 & -0.17 & 0.00 \\ -0.36 & -0.10 & -0.38 & 0.40 & 0.05 & -0.10 & 0.07 & 0.17 & -0.08 & 0.70 & 0.13 & 0.00 \\ 0.38 & 0.11 & 0.39 & -0.26 & -0.02 & 0.16 & -0.15 & -0.18 & 0.03 & 0.70 & -0.23 & 0.00 \\ -0.24 & -0.03 & -0.35 & -0.25 & 0.05 & 0.83 & -0.11 & -0.21 & -0.11 & -0.01 & 0.00 & 0.00 \end{bmatrix}_{12\times 12}$$

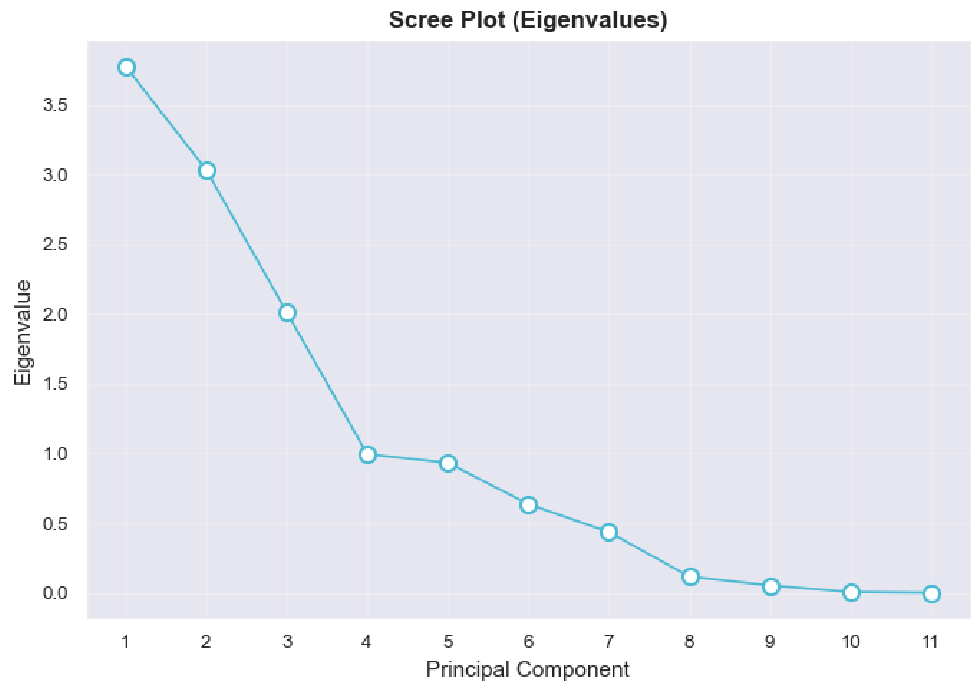
Rows: koi_period, koi_impact, koi_duration, koi_depth, koi_model_snr, koi_prad, koi_teq, koi_insol, koi_steff, koi_slogg, koi_srad, koi_kepmag
Columns: PC1 through PC12

Top Contributors by Component

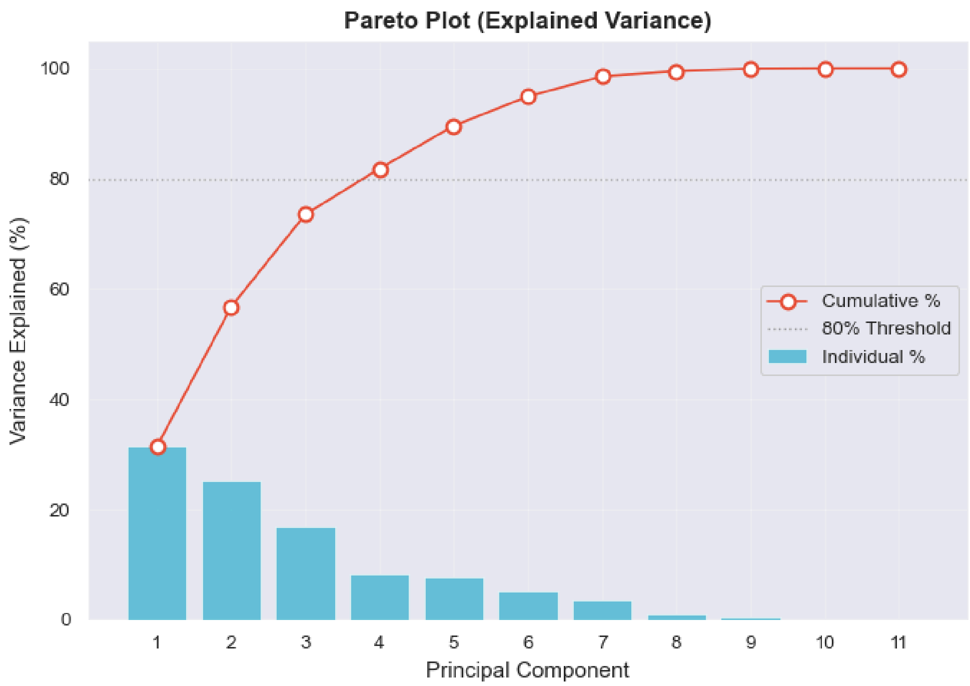
- **PC1** (31.14% variance): `koi_teq`, `koi_insol`, `koi_srad`, `koi_slogg`, `koi_period`
- **PC2** (24.76% variance): `koi_depth`, `koi_prad`, `koi_model_snr`, `koi_duration`, `koi_period`
- **PC3** (17.59% variance): `koi_srad`, `koi_period`, `koi_slogg`, `koi_kepmag`, `koi_depth`

Visualizations

Scree Plot and Pareto Plot (Cumulative Variance)

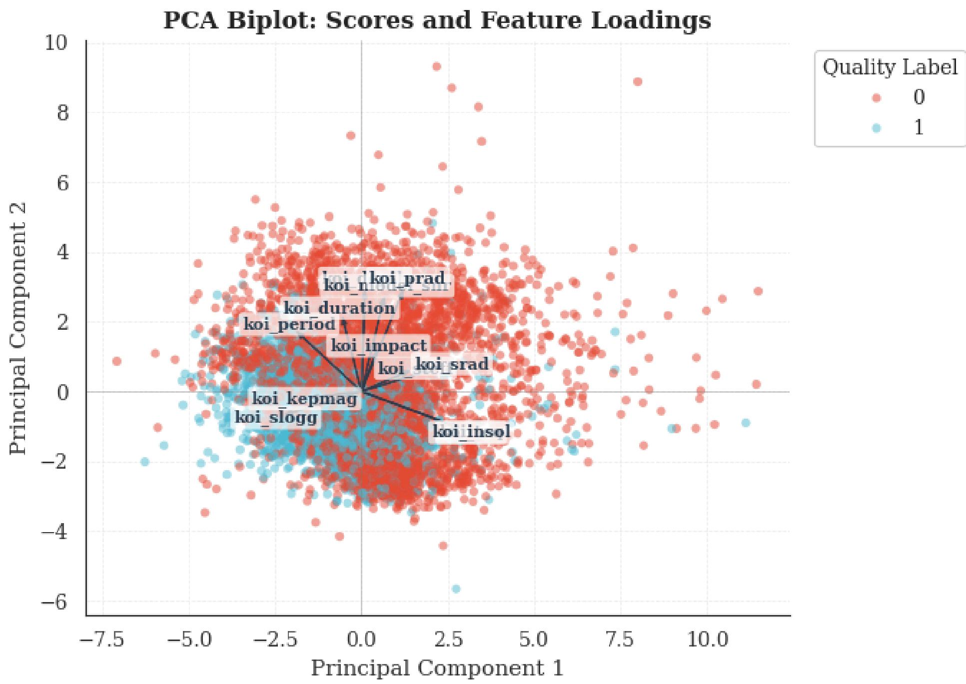


ScreePlot.png

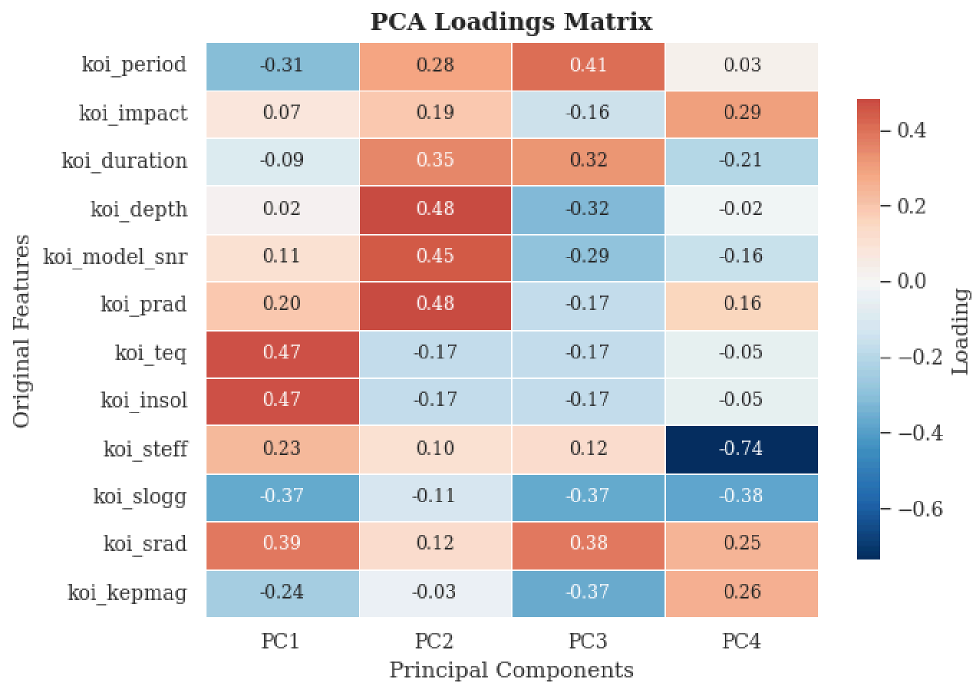


ParetoPlot.png

Biplot



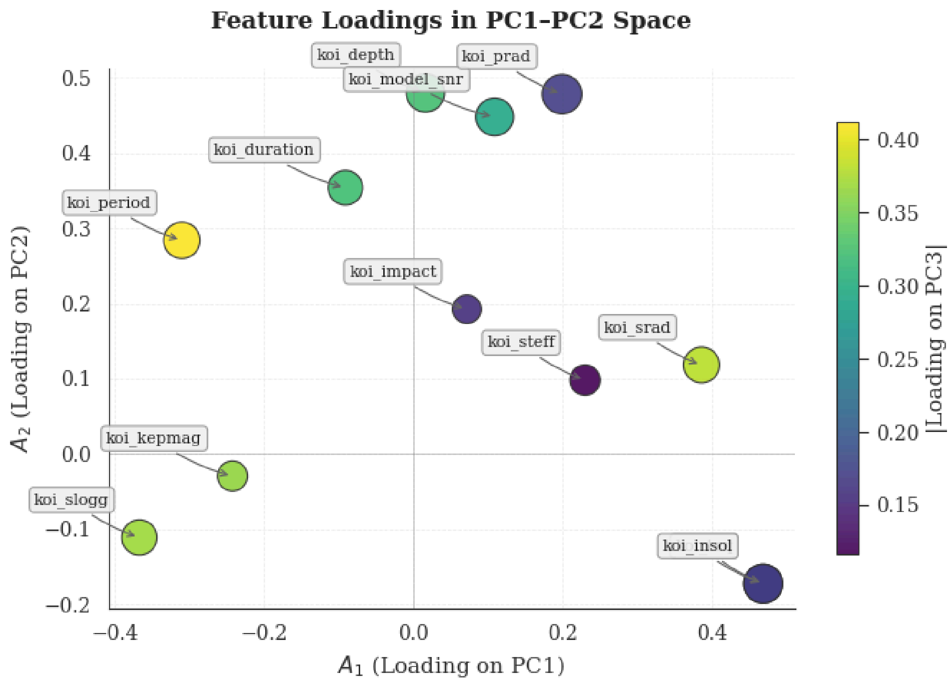
PCA_Biplot.png



	PC1	PC2	PC3	PC4
koi_period	-0.31	0.28	0.41	0.03
koi_impact	0.07	0.19	-0.16	0.29
koi_duration	-0.09	0.35	0.32	-0.21
koi_depth	0.02	0.48	-0.32	-0.02
koi_model_snr	0.11	0.45	-0.29	-0.16
koi_prad	0.20	0.48	-0.17	0.16
koi_teq	0.47	-0.17	-0.17	-0.05
koi_insol	0.47	-0.17	-0.17	-0.05
koi_steff	0.23	0.10	0.12	-0.74
koi_slogg	-0.37	-0.11	-0.37	-0.38
koi_srad	0.39	0.12	0.38	0.25
koi_kepmag	-0.24	-0.03	-0.37	0.26

PCA_Score_Plot.png

Loadings Heatmap



Feature_Loadings_Plot.png / PCA_Loading_Matrix.png

Model Comparison

Models on Original Features (12-D)

	Model	Accuracy	AUC	Recall	Prec.	F1	Kappa	MCC	TT (Sec)
lightgbm	Light Gradient Boosting Machine	0.8841	0.9509	0.9024	0.8627	0.8819	0.7682	0.7692	0.054
gbc	Gradient Boosting Classifier	0.8804	0.9445	0.9049	0.8544	0.8788	0.7609	0.7623	0.096
rf	Random Forest Classifier	0.8769	0.9440	0.8817	0.8646	0.8729	0.7535	0.7539	0.048
et	Extra Trees Classifier	0.8747	0.9428	0.8920	0.8535	0.8722	0.7493	0.7502	0.024
knn	K Neighbors Classifier	0.8677	0.9235	0.9003	0.8366	0.8672	0.7358	0.7380	0.011
ada	Ada Boost Classifier	0.8620	0.9245	0.8925	0.8322	0.8612	0.7244	0.7264	0.025
lda	Linear Discriminant Analysis	0.8058	0.8619	0.9334	0.7344	0.8218	0.6151	0.6372	0.003
ridge	Ridge Classifier	0.8055	0.8624	0.9329	0.7344	0.8216	0.6146	0.6366	0.002
lr	Logistic Regression	0.8137	0.8668	0.8889	0.7627	0.8207	0.6292	0.6380	0.004
qda	Quadratic Discriminant Analysis	0.7917	0.9003	0.9390	0.7158	0.8122	0.5877	0.6156	0.002
dt	Decision Tree Classifier	0.8164	0.8161	0.8088	0.8088	0.8083	0.6322	0.6329	0.007
svm	SVM - Linear Kernel	0.7902	0.8458	0.8435	0.7524	0.7940	0.5818	0.5882	0.005
nb	Naive Bayes	0.7756	0.8766	0.8977	0.7109	0.7933	0.5550	0.5735	0.004
dummy	Dummy Classifier	0.5204	0.5000	0.0000	0.0000	0.0000	0.0000	0.0000	0.002

Models on PCA Features (4 PCs)

	Model	Accuracy	AUC	Recall	Prec.	F1	Kappa	MCC	TT (Sec)
lightgbm	Light Gradient Boosting Machine	0.8063	0.8802	0.8274	0.7822	0.8039	0.6128	0.6143	0.042
rf	Random Forest Classifier	0.8093	0.8844	0.8099	0.7964	0.8028	0.6181	0.6187	0.040
gbc	Gradient Boosting Classifier	0.8023	0.8723	0.8243	0.7775	0.8000	0.6049	0.6063	0.043
et	Extra Trees Classifier	0.8036	0.8868	0.8130	0.7858	0.7989	0.6070	0.6077	0.025
knn	K Neighbors Classifier	0.8033	0.8704	0.8063	0.7894	0.7974	0.6064	0.6069	0.008
ada	Ada Boost Classifier	0.7865	0.8556	0.8239	0.7543	0.7874	0.5738	0.5763	0.017
qda	Quadratic Discriminant Analysis	0.7550	0.8594	0.9122	0.6833	0.7812	0.5156	0.5440	0.002
nb	Naive Bayes	0.7610	0.8463	0.8512	0.7092	0.7736	0.5249	0.5351	0.003
ridge	Ridge Classifier	0.7518	0.7956	0.8651	0.6937	0.7698	0.5075	0.5227	0.002
lda	Linear Discriminant Analysis	0.7518	0.7956	0.8651	0.6937	0.7698	0.5075	0.5227	0.002
lr	Logistic Regression	0.7473	0.7945	0.8248	0.7012	0.7578	0.4973	0.5050	0.003
svm	SVM - Linear Kernel	0.7406	0.7862	0.8124	0.6989	0.7484	0.4836	0.4952	0.004
dt	Decision Tree Classifier	0.7404	0.7399	0.7283	0.7308	0.7289	0.4799	0.4807	0.004
dummy	Dummy Classifier	0.5204	0.5000	0.0000	0.0000	0.0000	0.0000	0.0000	0.002

Model Training & Tuning

Training on Original Features

Logistic Regression (Original)

	Accuracy	AUC	Recall	Prec.	F1	Kappa	MCC
Mean	0.8134	0.8663	0.8889	0.7621	0.8204	0.6287	0.6375
Std	0.0126	0.0197	0.0238	0.0186	0.0116	0.0248	0.0243

Random Forest (Original)

	Accuracy	AUC	Recall	Prec.	F1	Kappa	MCC
Mean	0.8767	0.9440	0.8817	0.8644	0.8727	0.7533	0.7537
Std	0.0159	0.0107	0.0187	0.0185	0.0156	0.0317	0.0315

MLP Classifier (Original)

	Accuracy	AUC	Recall	Prec.	F1	Kappa	MCC
Mean	0.8913	0.9504	0.9027	0.8772	0.8897	0.7828	0.7830
Std	0.0130	0.0057	0.0161	0.0156	0.0132	0.0260	0.0261

Training on PCA Features

Logistic Regression (PCA)

	Accuracy	AUC	Recall	Prec.	F1	Kappa	MCC
Mean	0.7471	0.7943	0.8243	0.7014	0.7576	0.4968	0.5046
Std	0.0247	0.0196	0.0332	0.0239	0.0240	0.0491	0.0495

Random Forest (PCA)

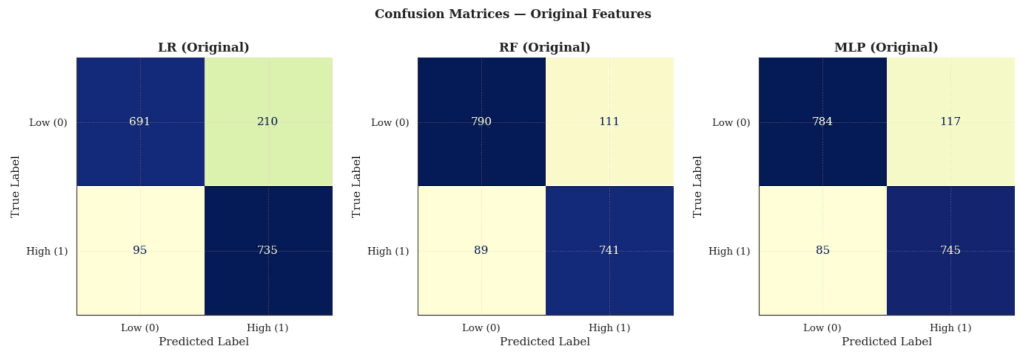
	Accuracy	AUC	Recall	Prec.	F1	Kappa	MCC
Mean	0.8089	0.8845	0.8100	0.7960	0.8024	0.6173	0.6179
Std	0.0162	0.0134	0.0213	0.0231	0.0153	0.0322	0.0315

MLP Classifier (PCA)

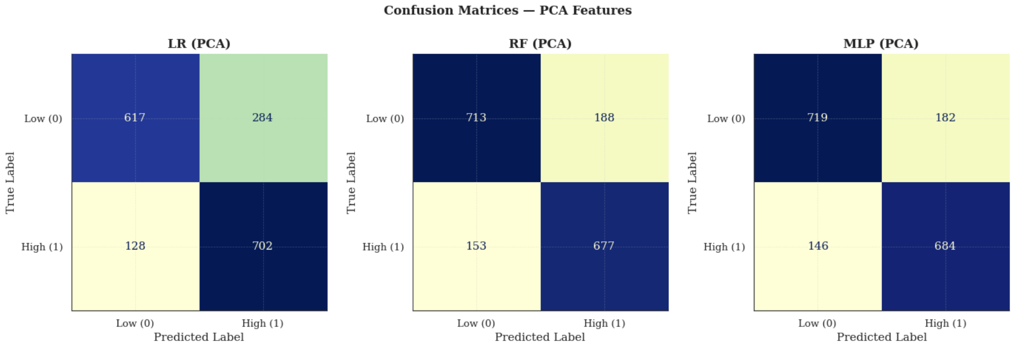
	Accuracy	AUC	Recall	Prec.	F1	Kappa	MCC
Mean	0.8099	0.8739	0.8049	0.8051	0.8044	0.6188	0.6185
Std	0.0119	0.0101	0.0142	0.0193	0.0103	0.0234	0.0229

Model Performance Visualization

Confusion Matrices

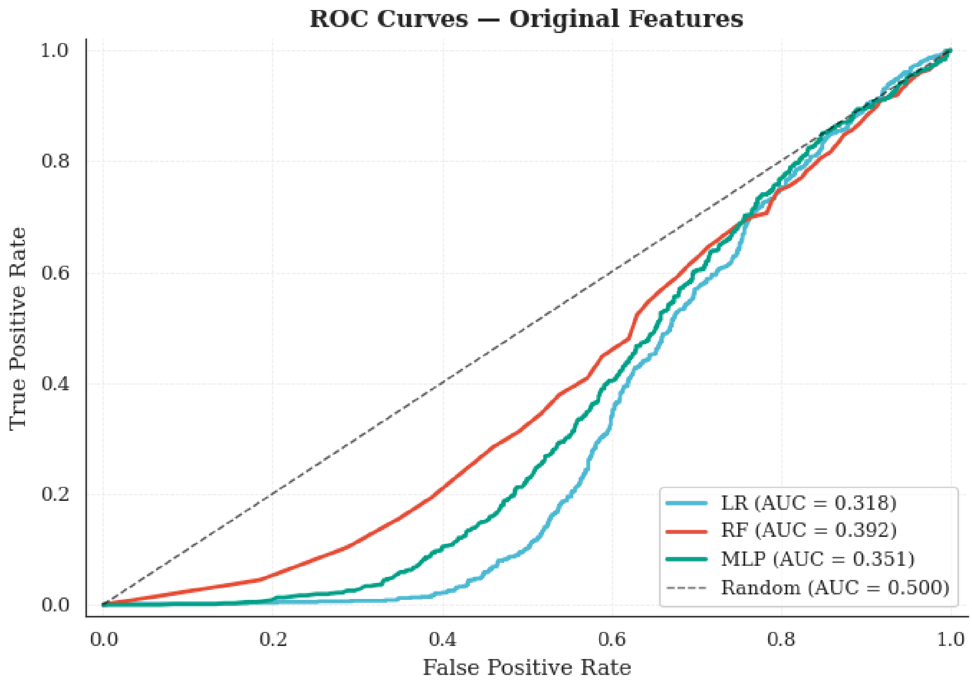


Confusion_Matrix_Original.png

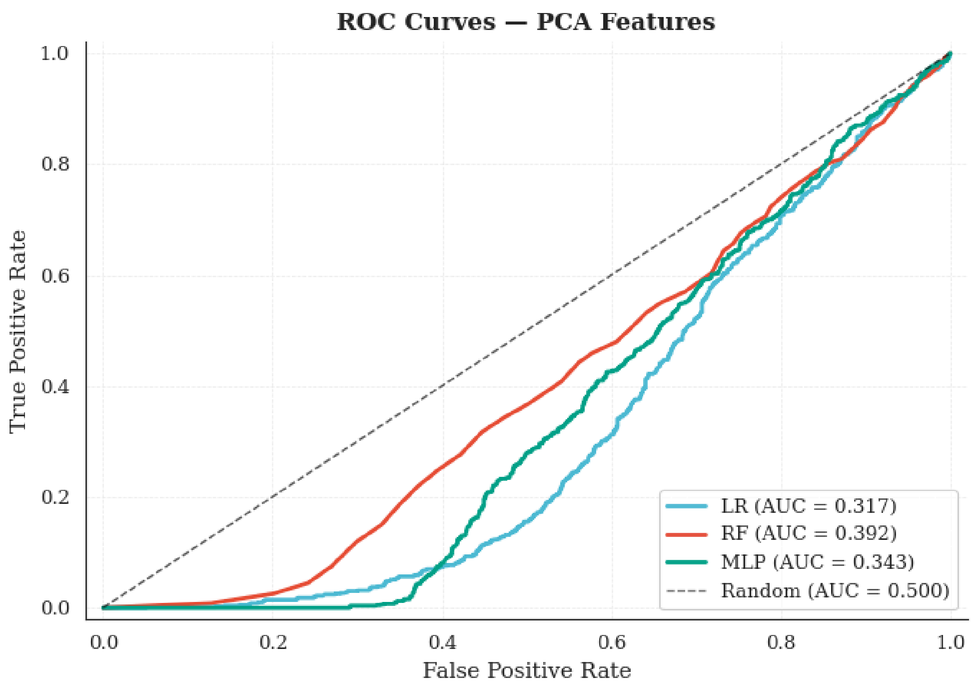


Confusion_Matrix_PCA.png

ROC Curves

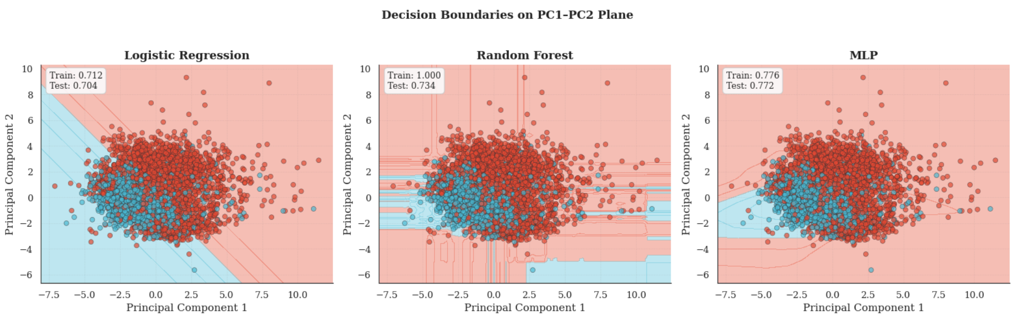


ROC_Curve_Original.png



ROC_Curve_PCA.png

Decision Boundaries on PC1-PC2 Plane



Decision_Boundaries.png

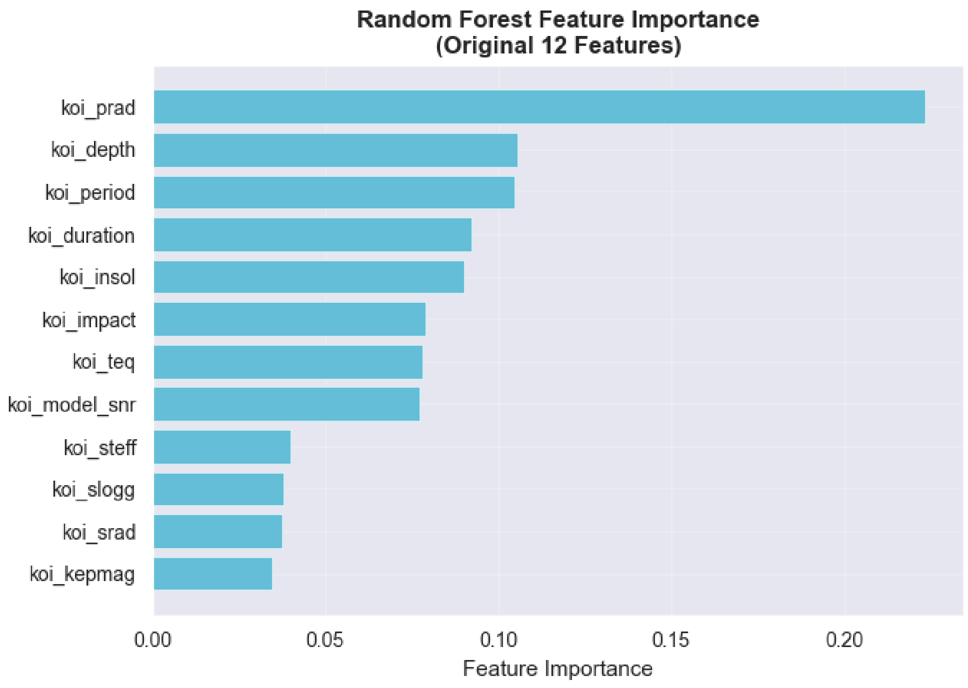
Final Model Comparison Summary

Test Set Performance

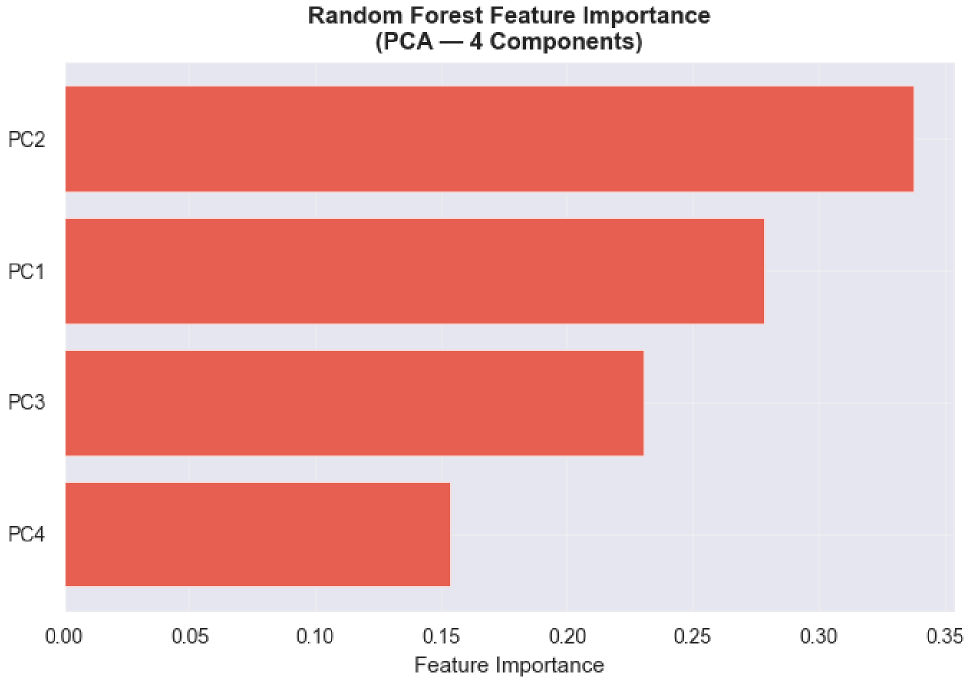
Model	Features	Accuracy	F1	Precision	Recall	AUC
LR	Original (12-D)	0.7990	0.8069	0.7481	0.8757	0.3651
LR	PCA (4 PCs)	0.7540	0.7669	0.7028	0.8439	0.3323
RF	Original (12-D)	0.8773	0.8738	0.8622	0.8858	0.3984
RF	PCA (4 PCs)	0.8011	0.7949	0.7864	0.8035	0.4178
MLP	Original (12-D)	**0.8926**	**0.8898**	0.8755	0.9046	0.3784
MLP	PCA (4 PCs)	0.8129	0.8049	0.8049	0.8049	0.3608

Best Model: MLP on Original (12-D) features
F1 Score: 0.8898 | **Accuracy:** 0.8926 | **AUC:** 0.3784

Feature Importance Analysis



RF_Feature_Importance_Original.png



RF_Feature_Importance_PCA.png

Top 5 Most Important Original Features

1. **koi_prad** — 0.2234
2. **koi_depth** — 0.1055

3. **koi_period** — 0.1045
4. **koi_duration** — 0.0921
5. **koi_insol** — 0.0902

Top 3 Most Important Principal Components

1. **PC2** — 0.3375
2. **PC1** — 0.2784
3. **PC3** — 0.2303